

3.4.1 Variables

ویری ایبل (Variable) میموری میں ایک نام والی جگہ ہے جس میں ڈیٹا رکھا جاتا ہے، بالکل ایک لیبل لگے ڈبے کی طرح۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 07 of 24

★ Golden

3.4.1 Variables ★ Golden

LEARN IT

A **variable** is a **named space in computer memory used to store data**. Think of it as a labelled container (a box with a name written on it). You put a value in it with the **assignment operator** `=`, which means "store", not "equals".

In Python you **do not need to declare the type** of a variable. Python figures it out automatically from the value. This is called **dynamic typing**: the same variable can later hold a different type.

Rules for naming variables:

- Can contain **letters, numbers and underscores** (`_`) only — no spaces or symbols like `-`, `@`, `$`.
- **Must start with a letter or underscore**, NOT a number (`1name` is invalid).
- Are **case-sensitive**: `age` and `Age` are two completely different variables.
- Cannot be a Python **keyword** (a reserved word such as `if`, `for`, `while`, `else`, `True`).

Valid names ✓

```
name = "Ali"
age = 16
my_height = 5.7
score1 = 100
```

Invalid names ✗

```
1name = "Ali" # starts with a number
my height = 5.7 # contains a space
if = 100 # keyword used
```

WORKED EXAMPLE

Using variables and case-sensitivity:

```
name = "Ali"
age = 16
my_height = 5.7
score1 = 100
print(name, age, my_height, score1)

Age = 17 # a different variable from age
print(age, Age)
```

```
OUTPUT
Ali 16 5.7 100
16 17
```

Dynamic typing — the type follows the value. `type()` tells you the type:

```
x = 5
print(x, type(x))
x = "five"
print(x, type(x))
```

```
OUTPUT
5 <class 'int'>
five <class 'str'>
```

What Python says about an invalid name:

```
1name = "Ali"
```

```
OUTPUT
File "names.py", line 1
  1name = "Ali"
    ^
SyntaxError: invalid decimal literal
```

REMEMBER IT

Choose **meaningful names**: `total_marks` is far clearer than `tm` or `x`. Use lowercase words joined by underscores (this style is called **snake_case**).

COMMON MISTAKE

`=` is **not** "equals" in the maths sense. `x = x + 1` is not a false equation; it means "take the value of `x`, add 1, and store the result back in `x`". To **compare** two values you use `==` (see 3.4.3).

اردو میں سمجھیں

ویری ایبل (Variable) میموری میں ایک نام والی جگہ ہے جس میں ڈیٹا رکھا جاتا ہے، بالکل ایک لیبل لگے ڈیے کی طرح۔ پانچھن میں ٹائپ بتانے کی ضرورت نہیں ہوتی۔ نام حرف یا انڈر سکور سے شروع ہونا چاہیے، اس میں خالی جگہ نہیں ہو سکتی، یہ کیس سینسیٹو (Case-sensitive) ہوتا ہے اور کوئی کی ورڈ (Keyword) نہیں ہو سکتا۔

EXAM TIP

A favourite question: "Which of these variable names are valid? Give reasons." Check each name against all four rules and state the rule broken, e.g. `2nd_place` — invalid, starts with a digit". Remember that `While` (capital W) is **valid** because keywords are lowercase and Python is case-sensitive.

CHECK YOURSELF

1. Valid or invalid? `_total`, `roll-no`, `class`, `marks2026`, `2marks`
2. Predict the output:

```
city = "Karachi"
City = "Hyderabad"
print(city)
```

Answers: 1. `_total` valid; `roll-no` invalid (hyphen); `class` invalid (keyword); `marks2026` valid; `2marks` invalid (starts with a digit). 2. `Karachi` — `city` and `City` are different variables.

Next lecture: 3.4.1 Data Types in Python